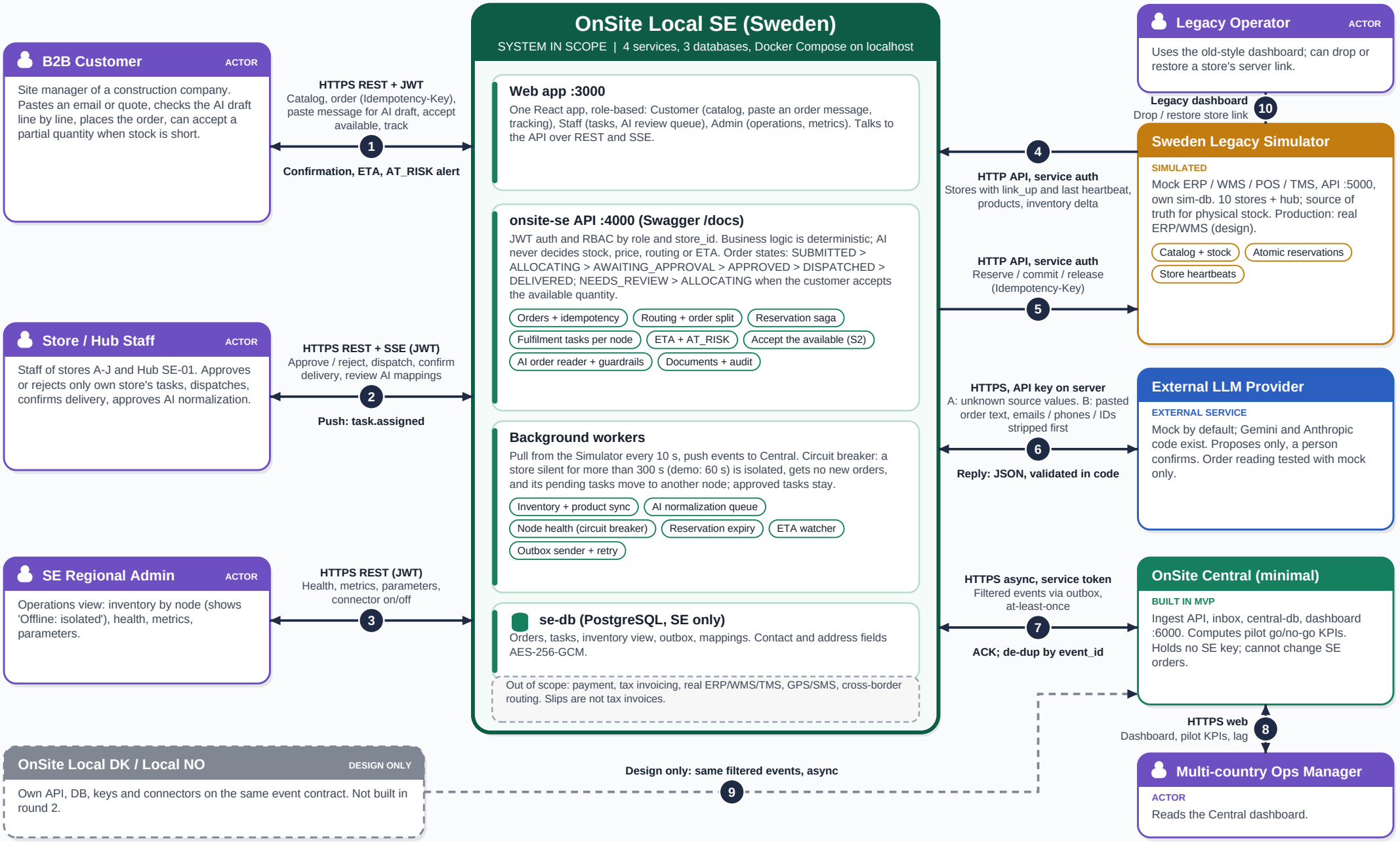
System Context Diagram: OnSite Local SE (Sweden MVP, Round 2)

Team PYRAMID | Shows only what the demo runs; simulated and design-only parts are labelled.

Human actor | Built and running in MVP | Simulated (real integration, mock data) | External service | Design only | Numbered badges = interactions, see page 2



Rules: sim-db, se-db and central-db are separate with no cross-DB reads. Local SE keeps working when Central is off; the Simulator runs when OnSite is off. Business figures are simulated.

Interaction catalogue and boundaries

Each interaction on page 1: protocol, data exchanged, what is deliberately not shared, and its status.

#	From - To	Protocol / auth	Data exchanged	Not shared / guardrails	Status
1	**B2B Customer <-> Local SE**	HTTPS REST via web UI. JWT, role customer.	Catalog; create order (Idempotency-Key); POST /orders/parse (text to draft, saves nothing); POST /orders/{id}/accept-available; status, ETA, late alerts.	Customer sees own orders only. 'Accept the 90 available' re-checks stock at click time, keeps requested_quantity (100) and is refused if nothing or a whole line is unavailable. Past delivery dates are rejected (DELIVERY_IN_PAST).	Built
2	**Store / Hub Staff <-> Local SE**	HTTPS REST + SSE. JWT, role staff, bound to store_id.	Task list, approve / reject, dispatch, delivery confirmation, AI review queue. Push: task.assigned.	store_id is checked on the server (403 + audit). SSE is a signal only; REST is the data source.	Built
3	**SE Regional Admin <-> Local SE**	HTTPS REST. JWT, role admin.	/health, /metrics, inventory by node with link state, parameters, connector on/off.	Cannot write data of other countries.	Built
4	**Simulator -> Local SE**	HTTP API, service credential. Polled every 10 s (cursor).	Stores incl. link_up and last_heartbeat_at, products (incl. new), inventory delta with updated_at and version.	No direct sim-db access. Node health (circuit breaker) uses the heartbeat time recorded by the source, not Local's clock, so a whole-source outage does not isolate every store. Stale data after 60 s is not committed on.	Simulated source, real integration
5	**Local SE -> Simulator**	HTTP API, service credential, Idempotency-Key.	Reserve (atomic, 30 min expiry), commit to stock-out, release; saga when several sources.	No customer data sent. Atomic reservation proven in simulation only. Offline store gets no new orders; its pending tasks move to another node, approved tasks stay.	Simulated source, real integration
6	**Local SE <-> External LLM**	HTTPS, API key on server only.	A) Normalization: unknown unit / description value, valid target list. B) Order reading: pasted text. Reply: JSON with value, quantity, quoted text, confidence.	Emails, phone numbers and personal IDs stripped in code before sending; no price or stock sent. Replies validated: verbatim quote, integer quantity 1 to 1,000,000, confidence below 0.7 flagged, unit mismatch (pallet vs sheet) flagged, only unflagged lines pre-ticked. AI never creates an order. Every suggestion waits for a person.	Built (mock tested; live providers untested)
7	**Local SE <-> OnSite Central**	HTTPS async, service token. Transactional outbox, retry.	Filtered events (e.g. order.approved.v1, task decisions, late alerts), then ACK.	No names, contacts, addresses or notes; no SE key at Central. Duplicates counted once. Optional demo delay is labelled 'simulated delay'.	Built (minimal)
8	**Ops Manager <-> Central**	HTTPS web dashboard.	Orders by status, AT_RISK, early shortages, sync lag, and 'Pilot go/no-go KPIs': OTIF, orders alerted before the customer had to ask, adoption.	Read-only. Alert count is not claimed as a measured drop in calls. Adoption turns to warning below 90%. OTIF is simulated data.	Built (minimal)
9	**Local DK / NO -> Central**	Same event contract as flow 7.	Same kind of filtered events.	Own DB, keys, connectors per country. Not built or tested.	Design only
10	**Legacy Operator <-> Simulator**	HTTP via the legacy dashboard UI.	Drop / Restore a store's server link; view stores, stock, orders.	Deliberately old-style UI to show before vs after OnSite. Demo uses a 60 s heartbeat timeout (default 300 s as in Round 1).	Simulated

Design rules behind the diagram

- OnSite is an API and orchestration layer on top of ERP/WMS/POS; it does not replace them (the Simulator stands in).
- Local SE is off Central's critical path: with Central down, orders continue and events replay later without duplicates.
- AI has two proposal-only roles: normalizing source data and reading a pasted order message. Stock, price, routing, reservation, ETA and tax stay deterministic.
- Round 1 mentioned auto-merge above 95% match; the MVP sends every AI suggestion to a person because accuracy is not yet measured.

Assumptions and limits (say them in the video)

- No real ERP/WMS/TMS: the Simulator has its own API and DB and is labelled simulated. Business numbers are simulated.
- Distances, speed, prep time, dispatch slots and the 60 s demo heartbeat timeout are configuration, not measurements.
- Round 1 KPIs (OTIF +3 pp, calls -30%, adoption >90%) are pilot targets, not results. AI accuracy is not measured.
- Keys and secrets come from env variables (no KMS); transport is plain localhost. Docker verified on Linux CI only.
- Denmark and Norway are design only; no cross-country routing.